

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (24 marks – approximately 43 minutes)

You are researching the movie theater industry in Country A. Specifically, you are assessing the potential impacts of two events that are taking place simultaneously in the country:

- Event (I) Int'lCin, an international film and television production and distribution company, is opening multi-screen movie theaters in most of the country's first-tier cities.
- Event (II) Movie theaters (existing and the new ones) are raising the price of snacks (such as popcorn and soft drinks) by an average of 30 percent.

Required:

- (a) Define substitutes and complements in consumption. (2 marks)
- (b) State any THREE factors that would generally cause:
- (i) a change in demand;
 - (ii) a change in supply. (3 marks)
- (c) Describe how the Latin phrase "ceteris paribus" (meaning all other factors being equal) can be used to explain the difference in "change in demand" and "change in quantity demanded". (2 marks)
- (d) Analyse how (the channel through which) Event (I) would affect the demand and/or supply of the movie tickets (seats). NO diagram is required. (2 marks)
- (e) Analyse how Event (II) would affect the demand and/or supply of the movie tickets. NO diagram is required. (3 marks)
- (f) Suppose you estimated the cross-price elasticity between movie tickets and snack price to be 0.7 (stated in ABSOLUTE value), analyse (with calculations support) how the movie ticket sales are affected by Event (II). (4 marks)
- (g) Analyse how the equilibrium price and quantity (sold) of movie tickets are affected by the impacts of the two events by filling in the following table:

	Direction of Price Change	Direction of Quantity Change
Event (I):	()	()
Event (II):	()	()
Net effect:	()	()

(8 marks)

Question 2 (24 marks – approximately 43 minutes)

Inflation and supply-chain disruptions are the buzzwords that have been dominating the news headlines in various countries over the past few months. Investors of all stripes have been paying close attention to how the unfolding of events would affect the world economy and the subsequent impacts brought about by government policy responses, especially those of central banks across the globe.

Required:

- (a) Define inflation, stagnation and stagflation. (4 marks)
- (b) (i) Justify why economists generally consider Consumer Price Index (CPI) an appropriate measure of consumers' cost of living. (2 marks)
- (ii) Describe the potential problem of CPI that may undermine its usefulness as a measure of cost of living for guiding government welfare policy. (2 marks)
- (c) Inflation is costly to society and the economy but does not affect every group of economic agents evenly.
- (i) Analyse the costs of inflation in terms of its redistribution effects, and deadweight loss effects. (4 marks)
- (ii) State and describe any ONE example of the redistributive effects of inflation. (2 marks)
- (d) The aggregate demand-aggregate supply (AD-AS) model is a macroeconomic model used to explain the relationship between price level and output of an economy.
- (i) Apply the AD-AS model to explain how supply chain disruption may result in stagflation. (2 marks)
- (ii) Explain the two types of inflation in an AD-AS context. (4 marks)
- (iii) In general, economists are pessimistic about the use of monetary policy in tackling stagflation. Use the AD-AS model to justify this view. (4 marks)

Question 3 (20 marks – approximately 36 minutes)

In a management context, statistics is a subject (a branch of mathematics) concerned with studying and developing methods for collecting, analysing, interpreting, and presenting data to facilitate decision making.

Required:

- (a) Explain what sampling is and how it is related to inferential statistics. (4 marks)
- (b) Continued with part (a). Analyse how statistical analysis can be used to assist decision making by stating the appropriate sequence with the terms provided below:
- () → () → () → () → () → ()
- Sampled data
 - Information
 - Raw data
 - ? sampling
 - Statistical inference
 - Population ?
- (4 marks)
- (c) Explain the key differences between time-series and cross-sectional data. (2 marks)
- (d) State and describe the FOUR components of time series. (6 marks)
- (e) Country A's economic activities have been severely disrupted since the beginning of 2020 by the lockdown measures brought about by the covid-19 pandemic. With an overwhelming percentage of the country's population having received full vaccination by the end of 2020, the country's tourism industry had gradually recovered since the first quarter of 2021. The following table shows the quarterly data (2020-21) on the dollar sales of the leisure and hospitalization sector.

Sales (Leisure & Hospitalisation Sector)	
Period	Sales (\$ billion)
Q1/2020	8
Q2/2020	4
Q3/2020	3
Q4/2020	2
Q1/2021	3
Q2/2021	4
Q3/2021	5
Q4/2021	6

Compute the quarterly sales growth rates of the last two quarters of 2021 on a year-over-year and quarter-over-quarter (i.e., sequential) basis.

(4 marks)

Question 4 (12 marks – approximately 22 minutes)

- (a) Define sampling error and confidence interval.

(3 marks)

- (b) Justify if the following statement is true or false:

"If the sample is small in size and its data points are spreading out, the sample mean is likely to be very close to the population mean."

(3 marks)

- (c) A professional examination tutoring company has conducted a survey to determine the proportion of its students who took a professional examination in the past two years. The survey covered 500 of the students, 480 of whom had taken the examination.

Required:

- (i) Compute a 95% confidence interval of the population proportion of students who took the examination in the past two years.

(4 marks)

- (ii) Determine whether the following statement is true or false.

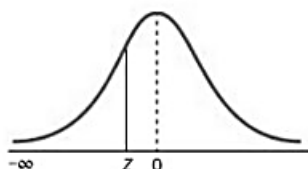
"A 95% confidence interval for the population mean implies that there is a 95% chance that the true population mean lies within the estimated interval computed based on a particular sample."

(2 marks)

* * * END OF EXAMINATION PAPER * * *

APPENDIX

The Cumulative Standardised Normal Distribution



Entry represents area under the cumulative standardised normal distribution from $-\infty$ to Z										
Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
-3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
-3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
-3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
-3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
-3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
-3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
-3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
-3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
-3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2388	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2482	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641